Top 10 GO Biological Process Gene Sets Ranked by Potency of Perturbation (Sorted by BMD 25% Relative Deviation at the 5th Percentile)

DTTID: 108-020-004-000-5
Study Number:108020004
Study Type: 5-Day Toxicity
Species/Strain: Rat HSD:Sprague Dawley SD

Test Compound: 1,4-Dichlorobenzene
CAS Number: 106-46-7
DTXSID: DTXSID1020431
Tissue: Liver

Date: 09 Jun 2025
Time: 09:35:54 AM

Female

Category Name	Input Genes/Platform Genes in Gene Set	% Gene Set Coverage	BMD_{rd25} at the 5th Percentile (ppm)	$BMDL_{rd25}$ at 5th Percentile - $BMDU_{rd25}$ at 5th Percentile (ppm)	Genes with Changed Direction Up	Genes with Changed Direction Down
GO:0006805 xenobiotic metabolic process	20/59	33.9	87.6	59.6 - 136.7	18	2
GO:0001676 long-chain fatty acid metabolic process	16/55	29.1	92.8	52.7 - 142.7	15	1
GO:0006690 icosanoid metabolic process	14/50	28.0	92.8	52.7 - 142.7	13	1
GO:0033559 unsaturated fatty acid metabolic process	16/54	29.6	92.8	52.7 - 142.7	15	1
GO:0120254 olefinic compound metabolic process	15/64	23.4	116.4	63.4 - 152.0	13	2
GO:0006720 isoprenoid metabolic process	17/41	41.5	143.6	107.8 - 165.0	15	2
GO:0071384 cellular response to corticosteroid stimulus	9/55	16.4	153.0	59.6 - 271.9	6	3
GO:0071385 cellular response to glucocorticoid stimulus	8/52	15.4	153.0	59.6 - 271.9	5	3
GO:0098754 detoxification	16/58	27.6	153.4	98.2 - 219.2	16	0
GO:0071383 cellular response to steroid hormone stimulus	10/62	16.1	158.6	44.1 - 339.0	6	4

rd25 = 25% relative deviation

Active Genes:
GO:0006805: Ugt2b1;Gstp1;Gstm2;Gsta4;Gsta1;Gsta3;Gsta2;Gsta5;Fmo1;Cyp3a2;Cyp3a23-3a1;Cyp2j4;Cyp2e1;Cyp2c6;Cyp2c11;Cyp2b2;Cyp2b1;Cbr1;Acaa1b;Acaa1a
GO:0001676: Mgll;Gstp1;Gstm2;Gsta1;Fads1;Ephx1;Elovl6;Cyp2j4;Cyp2e1;Cyp2c6;Cyp2c11;Cyp2b2;Cyp2b1;Cpt2;Acot4;Acadl
GO:0006690: Mif;Mgll;Gstp1;Gsta1;Gsta2;Fads1;Ephx1;Cyp2j4;Cyp2e1;Cyp2c6;Cyp2c11;Cyp2b2;Cyp2b1;Comt
GO:0033559: Mif;Mgll;Gstp1;Gstm2;Gsta1;Gsta2;Fads1;Ephx1;Elovl6;Cyp2j4;Cyp2e1;Cyp2c6;Cyp2c11;Cyp2b2;Cyp2b1;Comt
GO:0120254: Srd5a1;Mgll;Gstp1;Gstm2;Gsta1;Fads1;Ephx1;Cyp3a2;Cyp2j4;Cyp2e1;Cyp2c6;Cyp2c11;Cyp2b2;Cyp2b1;Aldh1a1
GO:0006720: Srd5a1;Pecr;Mvd;Lss;Idi1;Hmgcs1;Hmgcr;Fdps;Fdft1;Dhdds;Cyp3a2;Cyp2j4;Cyp2e1;Cyp2c6;Cyp2c11;Aldh1a1;Acat2
GO:0071384: Ugt2b1;Srd5a1;Ran;Pck1;Orm1;Gstp1;Ephx1;Eif4ebp1;Agtr1a
GO:0071385: Ugt2b1;Srd5a1;Pck1;Orm1;Gstp1;Ephx1;Eif4ebp1;Agtr1a
GO:0098754: Txnrd2;Txnrd1;Srxn1;Sod2;Slc22a1;Prdx3;Prdx2;Prdx1;Nqo1;Gstp1;Gstm2;Gsta3;Gsta2;Gsr;Aldh1a1;Akr7a3
GO:0071383: Ugt2b1;Srd5a1;Ran;Pck1;Orm1;Gstp1;Ephx1;Eif4ebp1;Egr1;Agtr1a

Annotations
Official active gene symbols from the Rat Genome Database are shown in the "Active Genes" section above. Definitions of Gene Ontology terms were provided by the Gene Ontology Resource (http://geneontology.org/) and are included below.
GO process description version: https://doi.org/10.22427/NTP-DATA-002-00600-0002-000-0 V06132025

GO Biological Process Definitions:
GO:0006805 xenobiotic metabolic process: The chemical reactions and pathways involving a xenobiotic compound, a compound foreign to living organisms. Used of chemical compounds, e.g. a xenobiotic chemical, such as a pesticide.
GO:0001676 long-chain fatty acid metabolic process: The chemical reactions and pathways involving long-chain fatty acids, A long-chain fatty acid is a fatty acid with a chain length between C13 and C22.
GO:0006690 icosanoid metabolic process: The chemical reactions and pathways involving icosanoids, any of a group of C20 polyunsaturated fatty acids.
GO:0033559 unsaturated fatty acid metabolic process: The chemical reactions and pathways involving an unsaturated fatty acid, any fatty acid containing one or more double bonds between carbon atoms.
GO:0120254 olefinic compound metabolic process: The chemical reactions and pathways involving an olefinic compound, any compound which contains a carbon-carbon double bond (aka C=C).
GO:0006720 isoprenoid metabolic process: The chemical reactions and pathways involving isoprenoid compounds, isoprene (2-methylbuta-1,3-diene) or compounds containing or derived from linked isoprene (3-methyl-2-butenylene) residues.

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GO:0071384 cellular response to corticosteroid stimulus: Any process that results in a change in state or activity of a cell (in terms of movement, secretion, enzyme production, gene expression, etc.) as a result of a corticosteroid hormone stimulus. A corticosteroid is a steroid hormone that is produced in the adrenal cortex. Corticosteroids are involved in a wide range of physiologic systems such as stress response, immune response and regulation of inflammation, carbohydrate metabolism, protein catabolism, blood electrolyte levels, and behavior. They include glucocorticoids and mineralocorticoids.

GO:0071385 cellular response to glucocorticoid stimulus: Any process that results in a change in state or activity of a cell (in terms of movement, secretion, enzyme production, gene expression, etc.) as a result of a glucocorticoid stimulus. Glucocorticoids are hormonal C21 corticosteroids synthesized from cholesterol with the ability to bind with the cortisol receptor and trigger similar effects. Glucocorticoids act primarily on carbohydrate and protein metabolism, and have anti-inflammatory effects.

GO:0098754 detoxification: Any process that reduces or removes the toxicity of a toxic substance. These may include transport of the toxic substance away from sensitive areas and to compartments or complexes whose purpose is sequestration of the toxic substance.

GO:0071383 cellular response to steroid hormone stimulus: Any process that results in a change in state or activity of a cell (in terms of movement, secretion, enzyme production, gene expression, etc.) as a result of a steroid hormone stimulus.

** END OF REPORT **